

Artificial Intelligence & Machine Learning - Task 3

Report

Title: Model Validation, Overfitting Control & Hyperparameter Tuning

Objective: Build a Decision Tree Regression model, evaluate it using train-test split and cross-validation, improve performance using GridSearchCV, and compare baseline and tuned models.

Tools: Python, NumPy, Pandas, scikit-learn, VS Code/Jupyter.

Dataset: California Housing dataset (20,640 samples, 8 features).

Methodology:

1. Load dataset.
2. Split into 80% training and 20% testing.
3. Train baseline DecisionTreeRegressor.
4. Evaluate using RMSE and R^2 .
5. Perform 5-fold cross-validation.
6. Tune hyperparameters using GridSearchCV.
7. Evaluate tuned model and compare results.

Results:

Baseline RMSE = 0.7030

Baseline R^2 = 0.6228

Cross-validation Average R^2 = 0.6070

Best Parameters: max_depth=10, min_samples_leaf=4, min_samples_split=2

Tuned RMSE = 0.6391

Tuned R^2 = 0.6883

Conclusion:

The tuned Decision Tree model outperformed the baseline model by reducing RMSE and increasing the R^2 score. Cross-validation confirmed consistent model performance, while GridSearchCV selected suitable hyperparameters that reduced overfitting and improved prediction accuracy.